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VEHICLE DRIVE SYSTEM

Abstract

A vehicle drive system including a motor, a temperature sensor, and a resolver includes a motor housing accommodating the motor, the temperature sensor and the resolver, and being open at one side in a motor axial direction, a cover member covering the opening, and temperature sensor and resolver harnesses connected in a first end part to the temperature sensor and the resolver, respectively. The cover member has a window for passing the temperature sensor harness. A temperature sensor connector connectable to a first external harness is attached to the cover member to penetrate a part thereof separated from the window. The temperature sensor connector is connected to a second end part of the temperature sensor harness. A resolver connector connectable to a second external harness is attached to and penetrates the motor housing. The resolver connector is connected to a second end part of the resolver harness.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a vehicle drive system, which is installed in a vehicle.

BACKGROUND OF THE DISCLOSURE

[0002] For example, JP7083298B discloses a vehicle provided with a motor for driving wheels. This motor is an in-wheel motor built into a wheel hub, provided with a hollow shaft that rotates with a rotor, and a sensor that detects a rotational position of the rotor. A harness of the sensor is inserted through the hollow shaft and extends to the outside of the motor to be connected to a control board.

[0003] Meanwhile, an internal temperature state and a rotational state of the motor are used as part of information when controlling the motor. In order to detect the internal temperature of the motor as accurately as possible, it is necessary to place a temperature sensor inside a housing accommodating the motor. Placing the temperature sensor inside the housing allows the sensor to be protected from an external environment, such as water and dust, which is thought to reduce issues like deterioration over time. Further, it is desired to place a resolver for detecting the rotational state of the motor inside the housing, in order to protect the resolver from the external environment.

[0004] When the temperature sensor and the resolver are placed inside the housing, harnesses extending from the temperature sensor and the resolver needs to be routed outside the housing. In this regard, JP7083298B routes the harness of the sensor through the hollow shaft to extend it outside. However, in a case where it is difficult to adopt such a hollow shaft structure, how to route the harness outside the housing becomes an issue. In particular, since the temperature sensor and the resolver are positioned at different locations, it may be challenging to adopt a layout where both the harness of the temperature sensor and the harness of the resolver pass through the hollow shaft as described in JP7083298B.

SUMMARY OF THE DISCLOSURE

[0005] The present disclosure is made in view of the above situations, and aims to output detection signals from a temperature sensor and a resolver to the outside of a housing, without adopting a hollow shaft structure, while avoiding localized concentration of a plurality of harnesses and facilitating a layout.

[0006] A vehicle drive system including a motor, a temperature sensor configured to detect a temperature state of the motor, and a resolver configured to detect a rotational state of the motor is provided, which includes a motor housing accommodating the motor, the temperature sensor, and the resolver, and being open at one side in an axial direction of a motor rotation shaft, a cover member covering the opening of the motor housing from the one side, a temperature sensor harness connected at a first end part to the temperature sensor, and a resolver harness connected at a first end part to the resolver. The cover member has a window through which the temperature sensor harness passes, and a temperature sensor connector connectable to a first external harness is attached to the cover member to penetrate a part of the cover member separated from the window. The temperature sensor connector is connected to a second end part of the temperature sensor harness. A resolver connector connectable to a second external harness is attached to the motor housing to penetrate the motor housing. The resolver connector is connected to a second end part of the resolver harness.

[0007] The cover member may include a cover-side vertical wall part extending in a radial

direction of the motor rotation shaft to cover the opening of the motor housing, and a cover-side circumferential wall part projecting in the axial direction of the motor rotation shaft from a circumferential edge portion of the cover-side vertical wall part to extend annularly. The window may be formed in the cover-side vertical wall part. The temperature sensor connector may be attached to the cover-side circumferential wall part to penetrate the cover-side circumferential wall part.

[0008] The temperature sensor and the resolver may be disposed away from each other in the axial direction of the motor rotation shaft.

[0009] The temperature sensor may be accommodated inside the motor housing on a side closer to the cover-side vertical wall part.

[0010] The resolver may be accommodated inside the motor housing on an opposite side from the cover-side vertical wall part.

[0011] The motor housing may have a housing-side vertical wall part extending in the radial direction of the motor rotation shaft, and a housing-side circumferential wall part projecting in the axial direction of the motor rotation shaft from a circumferential edge portion of the housing-side vertical wall part to extend annularly, and coupled to the cover member. The resolver connector may be attached to the housing-side circumferential wall part to penetrate the housing-side circumferential wall part.

[0012] A side of the temperature sensor connector to which the first external harness is connected may face the same direction as a side of the resolver connector to which the second external harness is connected.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. 1 is a frontal perspective view of a vehicle drive system.

[0014] FIG. 2 is a front view of the vehicle drive system.

[0015] FIG. 3 is a schematic cross-sectional view taken along a line III-III in FIG. 2.

[0016] FIG. 4 is an upper-left perspective view illustrating a separated state of a right cover member and a motor housing.

[0017] FIG. 5 is an upper-right perspective view illustrating the separated state of the right cover member and the motor housing.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0018] Hereinafter, one embodiment of the present disclosure will be described in detail with reference to the accompanying drawings. Note that the following description of this embodiment is merely illustrative in nature and is not intended to limit the present disclosure, its applications, or uses.

[0019] FIGS. 1 and 2 illustrate a vehicle drive system 1 according to an embodiment of the present disclosure. The vehicle drive system 1 is installed in an automobile (not illustrated) and drives drive wheels of the automobile. The automobile may be an electric vehicle which runs solely on power of a motor or a hybrid vehicle which uses both an internal combustion engine (engine) and the motor. The motor for generating the power to drive the drive wheels is a drive motor. The hybrid automobile may be a plug-in hybrid which is chargeable from an external power source. In this embodiment, the explanation is given for a case where a vehicle on which the vehicle drive system 1 is installed is the hybrid vehicle. Therefore, as illustrated in FIG. 3, although the automobile on which the vehicle drive system 1 is installed is also equipped with a drive battery 100 which supplies power to a motor 2 and a controller 200 which controls the motor 2, since they are conventionally well-known, the detailed description thereof is omitted. The controller 200 includes, for example, an inverter circuit.

[0020] The vehicle drive system **1** is mounted, for example, in an engine compartment of the automobile. The engine compartment may be located in a front or rear part of the automobile. Front, rear, left, and right directions are defined as illustrated in each drawing. That is, with reference to the state where the vehicle drive system **1** is mounted in the engine compartment of the automobile, the front side of the vehicle is simply referred to as “front,” the rear side of the vehicle is simply referred to as “rear,” the left side of the vehicle is simply referred to as “left,” and the right side of the vehicle is simply referred to as “right.” A left-and-right direction of the vehicle corresponds to a width direction of the vehicle. This definition of the directions is solely for the convenience in describing the embodiment and does not limit the present disclosure.

[0021] As partially illustrated by virtual lines in FIG. 2, an engine E is disposed inside the engine compartment, on the right side of the vehicle drive system **1**. A crankshaft (not illustrated) of the engine E extends in the left-and-right direction, that is, the engine E is a so-called transverse engine. The engine E and the vehicle drive system **1** are arranged side by side in the left-and-right direction, the vehicle drive system **1** is disposed on the left side of the engine E, and a left part of the engine E is coupled to a right part of the vehicle drive system **1** by a fastening member (not illustrated). The power outputted from the crankshaft of the engine E is inputted to the vehicle drive system **1**. Note that the left and right positional relationship may be reversed, and the arranged direction of the engine E and the vehicle drive system **1** is not particularly restricted. In the case of an electric vehicle, since the engine E is not installed, the engine compartment is referred to as a motor compartment.

[0022] FIG. 3 schematically illustrates a cross section of the vehicle drive system **1**. As illustrated in FIG. 3, the vehicle drive system **1** includes the motor **2**, a temperature sensor **3** which detects a temperature of the motor **2**, and a resolver **4** which detects a rotational state of the motor **2**. As illustrated in FIG. 4, the motor **2** includes a stator **2a** and a rotor **2b** (only an outer circumference thereof is partially illustrated by a virtual line). The core of a rotation shaft of the motor **2** (motor rotation shaft) is denoted by a reference character B in FIG. 2. A mounting direction of the vehicle drive system **1** in the vehicle is set so that a motor rotation shaft direction aligns with the left-and-right direction. Therefore, an axis core direction of the crankshaft of the engine E matches the motor rotation shaft direction.

[0023] The temperature sensor **3** is, for example, comprised of a thermistor and is disposed close to the stator **2a** of the motor **2** in order to immediately detect a temperature change in the stator **2a**. The temperature sensor **3** constantly outputs a detection signal. The resolver **4** is a rotational angle detection sensor which detects a rotational angle of the rotor **2b**. The temperature sensor **3** and the resolver **4** may be of any commonly known type.

[0024] The vehicle drive system **1** also includes a motor housing **10**, a right cover member **20**, and a left cover member **30**. The motor housing **10**, the right cover member **20**, and the left cover member **30** are each made of lightweight alloy, such as an aluminum alloy, providing high rigidity. Although in this embodiment the right and left cover members **20** and **30** are named this way since they are disposed on the respective right and left sides, without limiting to this, the right cover member **20** may be referred to as a front cover member and the left cover member **30** as a rear cover member. The right cover member **20** is a first cover member, and the left cover member **30** is a second cover member.

[0025] The motor housing **10**, the right cover member **20**, and the left cover member **30** form a single case A. The right cover member **20** is positioned on the right side of the case A to constitute a right part of the case A, and thus may also be referred to as a “right-side component.” The left cover member **30** is positioned on the left side of the case A to constitute a left part of the case A, and thus may also be referred to as a “left-side component.” The motor housing **10** is positioned between the right and left cover members **20** and **30** to constitute an intermediate part of the case A in the left-and-right direction, and thus may also be referred to as an “intermediate component.” The motor housing **10** is sandwiched in the left-and-right direction by the right and left cover

members **20** and **30**. The structure of the case A is not limited to the above structure and may include additional component(s).

[0026] The motor housing **10** forms a single accommodation space R (illustrated in FIG. 3). The motor **2**, the temperature sensor **3**, and the resolver **4** are accommodated inside the single accommodation space R. The motor housing **10** includes a housing-side vertical wall part **11** and a housing-side circumferential wall part **12**, which are integrally molded. The housing-side vertical wall part **11** is positioned at a left end part of the motor housing **10** and is an end wall part extending in a radial direction (up-and-down and front-and-rear directions) of the motor rotation shaft. The housing-side circumferential wall part **12** projects rightward along the motor rotation shaft direction from a circumferential edge portion of the housing-side vertical wall part **11** and extends continuously in a circumferential direction of the motor rotation shaft to form an annular wall part. In a tip end portion (right end portion) of the housing-side circumferential wall part **12**, a housing-side fastening portion **12a** protruding radially outward is formed. The housing-side fastening portion **12a** is joined to the right cover member **20** by a fastening member (not illustrated).

[0027] The motor **2** is positioned inside the housing-side circumferential wall part **12**. An output combining mechanism (not illustrated) for combining the output of the engine E with the output of the motor **2** (rotational force of the rotor **2b**) is provided inside the rotor **2b** (illustrated by virtual lines in FIG. 4). The output combining mechanism is capable of inputting only the output of the engine E, only the output of the motor **2**, or a combination of the outputs of the engine E and the motor **2** at an arbitrary ratio, to a subsequent transmission **50** (illustrated in FIG. 3). The structure of the output combining mechanism is not particularly limited and may apply a conventionally well-known mechanism used in hybrid vehicles. The control of the output combining mechanism may also use a conventionally well-known method.

[0028] The housing-side vertical wall part **11** is located in the left end part of the motor housing **10**, as the end wall part. However, since no end wall part is formed in the right end part of the motor housing **10**, the motor housing **10** has a structure where it has an opening formed in the right direction (one side in the motor rotation shaft direction). Since the accommodation space R opens in the right direction of the motor housing **10**, the opening of the motor housing **10** is covered by the right cover member **20** from the right side.

[0029] The right cover member **20** includes a right cover-side vertical wall part **21** and a right cover-side circumferential wall part **22**, which are integrally molded. The right cover-side vertical wall part **21** is an end wall part located in a left end part of the right cover member **20** and extending in the radial direction (up-and-down and front-and-rear directions) of the motor rotation shaft to cover the opening of the motor housing **10**. The housing-side fastening portion **12a** is joined to the right cover-side vertical wall part **21** to close the opening of the motor housing **10**.

[0030] The right cover-side circumferential wall part **22** projects from a circumferential edge portion of the right cover-side vertical wall part **21** in the motor rotation shaft direction and extends continuously in the circumferential direction of the motor rotation shaft to form an annular wall part. In a tip end portion (right end portion) of the right cover-side circumferential wall part **22**, a right cover-side fastening portion **22a** protruding radially outward is formed. The right cover-side fastening portion **22a** is connected to the engine E by a fastening member (not illustrated).

[0031] The left cover member **30** is attached to the motor housing **10** at the opposite side (left side) from the opening and includes a left cover-side vertical wall part **31** and a left cover-side circumferential wall part **32**, which are integrally molded. The left cover-side vertical wall part **31** is an end wall part located in a left end part of the left cover member **30** and extending in the radial direction (up-and-down and front-and-rear directions) of the motor rotation shaft.

[0032] The left cover-side circumferential wall part **32** projects from a circumferential edge portion of the left cover-side vertical wall part **31** in the motor rotation shaft direction and extends continuously in the circumferential direction of the motor rotation shaft to form an annular wall

part. In a tip end portion (right end portion) of the left cover-side circumferential wall part **32**, a left cover-side fastening portion **32a** protruding radially outward is formed. The left cover-side fastening portion **32a** is coupled to the housing-side vertical wall part **11** of the motor housing **10** by a fastening member (not illustrated).

[0033] The transmission **50** is accommodated inside the left cover-side circumferential wall part **32**. The rotational force outputted from the output combining mechanism accommodated in the motor housing **10** is inputted to the transmission **50** to be decelerated or accelerated and then be outputted. The output of the transmission **50** is transmitted to the drive wheels via a final gear and a drive shaft (not illustrated).

[0034] The temperature sensor **3** and the resolver **4** are spaced apart in the motor rotation shaft direction. The temperature sensor **3** is accommodated inside the motor housing **10** near the right cover-side vertical wall part **21** and, in this embodiment, is fixed by being fastened to the right cover-side vertical wall part **21**. The resolver **4** is accommodated inside the motor housing **10** on the opposite side from the right cover-side vertical wall part **21** and is fixed by being fastened to the housing-side vertical wall part **11**. Thus, the temperature sensor **3** is accommodated on the right side of the accommodation space R and the resolver **4** is accommodated on the left side of the accommodation space R.

[0035] The vehicle drive system **1** further includes a temperature sensor harness **3a** connected at its base end part (first end part) to the temperature sensor **3**, and a resolver harness **4a** connected at its base end part (first end part) to the resolver **4**. The temperature sensor harness **3a** contains a signal line which transmits the detection signal of the temperature sensor **3** to the controller **200** located outside the case A. The resolver harness **4a** contains a signal line which transmits the detection signal of the resolver **4** to the controller **200**.

[0036] The right cover-side vertical wall part **21** of the right cover member **20** has a window **21a** through which the temperature sensor harness **3a** passes. The window **21a** is located radially outward of the stator **2a** in the right cover-side vertical wall part **21** and is formed to extend through the right cover-side vertical wall part **21** in the thickness direction (left-and-right direction). In this embodiment, the window **21a** is located in a front portion of the right cover-side vertical wall part **21**. The formation of the window **21a** allows the temperature sensor harness **3a** to be routed from the inside of the motor housing **10** to the inside of the right cover member **20**. An inner surface of the window **21a** and the temperature sensor harness **3a** can be sealed therebetween with a sealing material.

[0037] A temperature sensor connector **3b** is attached to the right cover-side circumferential wall part **22** of the right cover member **20**. Specifically, a connector attachment hole **22b** passing through the right cover-side circumferential wall part **22** in the thickness direction is formed in a front portion of the right cover-side circumferential wall part **22**. The temperature sensor connector **3b** is fixed to the right cover-side circumferential wall part **22** in a state inserted into the connector attachment hole **22b**. In other words, the temperature sensor connector **3b** is attached to the right cover-side circumferential wall part **22**, which is separated from the window **21a**, to pass therethrough.

[0038] A tip end part (second end part) of the temperature sensor harness **3a** is connected to the temperature sensor connector **3b**. The temperature sensor connector **3b** is connectable to a first external harness **301**. The side of the temperature sensor connector **3b** connected to the first external harness **301** faces forward.

[0039] The first external harness **301** includes a signal line for transmitting the detection signal of the temperature sensor **3** to the controller **200**. An end part of the first external harness **301** is equipped with a first connector **302** connected to the temperature sensor connector **3b**. Since a front part of the temperature sensor connector **3b** is connected to the first connector **302**, a rear part of the first connector **302** is connected to the temperature sensor connector **3b**. One of the temperature sensor connector **3b** and the first connector **302** may be a female connector or a male connector.

[0040] For example, one of the temperature sensor connector **3b** and the first connector **302** may have an insertion port to be a female connector, and the other one of the temperature sensor connector **3b** and the first connector **302** may have an inserting part to be inserted into the insertion port, to be the male connector. For example, when the temperature sensor connector **3b** has the insertion port, the arrangement of the temperature sensor connector **3b** may be determined so that the insertion port faces forward. In this case, by inserting the inserting part, which is formed in the first connector **302**, into the insertion port of the temperature sensor connector **3b** from its front side, the first connector **302** can be connected to the temperature sensor connector **3b**. Conversely, when the inserting part is formed in the temperature sensor connector **3b**, the arrangement of the temperature sensor connector **3b** may be determined so that the inserting part faces forward.

[0041] A resolver connector **4b** configured such that a second external harness **303** is connectable is attached to the motor housing **10** to pass therethrough. Specifically, the housing-side circumferential wall part **12** has a connector attachment hole **12b** extending through the housing-side circumferential wall part **12** in the thickness (radial) direction. The resolver connector **4b** is fixed to the housing-side circumferential wall part **12** in a state where it is inserted into the connector attachment hole **12b**. In other words, the resolver connector **4b** is attached to the motor housing **10** to pass through a position thereof separated from the temperature sensor connector **3b** in the left-and-right direction.

[0042] A tip end part (second end part) of the resolver harness **4a** is connected to the resolver connector **4b**. The resolver connector **4b** is connectable to the second external harness **303**. The side of the resolver connector **4b** connected to the second external harness **303** faces forward. Thus, the side of the temperature sensor connector **3b** where the first external harness **301** connects and the side of the resolver connector **4b** where the second external harness **303** connects both face the same direction. These connection sides may alternatively face upward, downward, or rearward.

[0043] The second external harness **303** contains a signal line for transmitting the detection signal of the resolver **4** to the controller **200**. A second connector **304** connected to the resolver connector **4b** is provided in an end part of the second external harness **303**. Since the second connector **304** is connected to a front part of the resolver connector **4b**, a rear part of the second connector **304** is connected to the resolver connector **4b**. Either of the resolver connector **4b** and the second connector **304** may be the female or male connector. Similarly to the temperature sensor connector **3b** and the first connector **302**, for example, one of the resolver connector **4b** and the second connector **304** may have an insertion port to be a female connector, and the other one of the resolver connector **4b** and the second connector **304** may have an inserting part to be a male connector. The first external harness **301** and the second external harness **303** may be bundled into a single harness outside the case A.

Operation and Effect(s) of Embodiment

[0044] As described above, according to this embodiment, since the temperature sensor **3** for detecting the temperature state of the motor **2** is provided inside the motor housing **10**, the temperature state of the motor **2** may accurately be detected. The temperature sensor harness **3a** connected to the temperature sensor **3** is connected to the temperature sensor connector **3b** attached to the right cover member **20**, while passing through the window **21a** of the right cover member **20**. Since the first external harness **301** is connected to the temperature sensor connector **3b**, the detection signal from the temperature sensor **3** is inputted to the controller **200** via the temperature sensor harness **3a**, the temperature sensor connector **3b**, and the first external harness **301**.

[0045] Further, the resolver harness **4a** connected to the resolver **4** is connected to the resolver connector **4b** attached to the motor housing **10**. Since the second external harness **303** is connected to the resolver connector **4b**, the detection signal of the resolver **4** can be inputted to the controller **200**, for example, via the resolver harness **4a**, the resolver connector **4b**, and the second external harness **303**.

[0046] Thus, by providing the temperature sensor connector **3b** and the resolver connector **4b**, it

becomes possible to input the detection signals of the temperature sensor **3** and the resolver **4** to the controller **200** without adopting the hollow shaft structure as in the conventional example. Further, since the temperature sensor connector **3b** and the resolver connector **4b** penetrate the right cover member **20** and the motor housing **10**, respectively, the connectors **3b** and **4b** and the harnesses **3a** and **4a** are not locally concentrated, making it easier to facilitate layout, as compared to a case where the connectors **3b** and **4b** are attached through a single member. Furthermore, there is no need to form through-hole(s) in the left cover member **30** for the connectors **3b** and **4b**.

[0047] The above embodiment is merely illustrative in every aspect and must not be interpreted restrictively. Moreover, any modifications and/or changes within the scope of equivalency of the appended claims are within the scope of the present disclosure.

INDUSTRIAL APPLICABILITY

[0048] As described above, the vehicle drive system according to the present disclosure can be used in hybrid automobiles or electric automobiles, for example.

[0049] It should be understood that the embodiments herein are illustrative and not restrictive, since the scope of the invention is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof, are therefore intended to be embraced by the claims.

DESCRIPTION OF REFERENCE CHARACTERS

[0050] **1** Vehicle Drive System [0051] **2** Motor [0052] **3** Temperature Sensor [0053] **3a** Temperature Sensor Harness [0054] **3b** Temperature Sensor Connector [0055] **4** Resolver [0056] **4a** Resolver Harness [0057] **4b** Resolver Connector [0058] **10** Motor Housing [0059] **20** Right Cover Member [0060] **21** Right Cover-side Vertical Wall Part [0061] **21a** Window [0062] **22** Right Cover-side Circumferential Wall Part [0063] **301** First External Harness [0064] **303** Second External Harness

Claims

1. A vehicle drive system including a motor, a temperature sensor configured to detect a temperature state of the motor, and a resolver configured to detect a rotational state of the motor, further comprising: a motor housing accommodating the motor, the temperature sensor, and the resolver, and being open at one side in an axial direction of a motor rotation shaft; a cover member covering the opening of the motor housing from the one side; a temperature sensor harness connected at a first end part thereof to the temperature sensor; and a resolver harness connected at a first end part thereof to the resolver, wherein the cover member has a window through which the temperature sensor harness passes, and a temperature sensor connector connectable to a first external harness is attached to the cover member to penetrate a part of the cover member separated from the window, wherein the temperature sensor connector is connected to a second end part of the temperature sensor harness, wherein a resolver connector connectable to a second external harness is attached to the motor housing to penetrate the motor housing, and wherein the resolver connector is connected to a second end part of the resolver harness.
2. The vehicle drive system of claim 1, wherein the cover member includes a cover-side vertical wall part extending in a radial direction of the motor rotation shaft to cover the opening of the motor housing, and a cover-side circumferential wall part projecting in the axial direction of the motor rotation shaft from a circumferential edge portion of the cover-side vertical wall part to extend annularly, wherein the window is formed in the cover-side vertical wall part, and wherein the temperature sensor connector is attached to the cover-side circumferential wall part to penetrate the cover-side circumferential wall part.
3. The vehicle drive system of claim 2, wherein the temperature sensor and the resolver are disposed away from each other in the axial direction of the motor rotation shaft.
4. The vehicle drive system of claim 3, wherein the temperature sensor is accommodated inside the

motor housing on a side closer to the cover-side vertical wall part.

5. The vehicle drive system of claim 4, wherein the resolver is accommodated inside the motor housing on an opposite side from the cover-side vertical wall part.

6. The vehicle drive system of claim 5, wherein the motor housing has a housing-side vertical wall part extending in the radial direction of the motor rotation shaft, and a housing-side circumferential wall part projecting in the axial direction of the motor rotation shaft from a circumferential edge portion of the housing-side vertical wall part to extend annularly, and coupled to the cover member, and wherein the resolver connector is attached to the housing-side circumferential wall part to penetrate the housing-side circumferential wall part.

7. The vehicle drive system of claim 6, wherein a side of the temperature sensor connector to which the first external harness is connected faces the same direction as a side of the resolver connector to which the second external harness is connected.

8. The vehicle drive system of claim 1, wherein the temperature sensor and the resolver are disposed away from each other in the axial direction of the motor rotation shaft.

9. The vehicle drive system of claim 8, wherein the temperature sensor is accommodated inside the motor housing on a side closer to the cover-side vertical wall part.

10. The vehicle drive system of claim 9, wherein the resolver is accommodated inside the motor housing on an opposite side from the cover-side vertical wall part.

11. The vehicle drive system of claim 10, wherein the motor housing has a housing-side vertical wall part extending in the radial direction of the motor rotation shaft, and a housing-side circumferential wall part projecting in the axial direction of the motor rotation shaft from a circumferential edge portion of the housing-side vertical wall part to extend annularly, and coupled to the cover member, and wherein the resolver connector is attached to the housing-side circumferential wall part to penetrate the housing-side circumferential wall part.

12. The vehicle drive system of claim 11, wherein a side of the temperature sensor connector to which the first external harness is connected faces the same direction as a side of the resolver connector to which the second external harness is connected.
